

Lattice-Convex Realization and Rank-Raising Topology of Optimal Completion Fibers

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Abstract

Let $A \in \mathbb{Z}^{k \times d}$ and let r be a completable residual. The minimum-completion problem

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{Z}_{\geq 0}^d, Ax = -r\}$$

has a finite optimal Parikh fiber $\mathcal{O}_A(r)$. Previous completion geometry reduces the filled shortest-prefix quotient to a periodic translation quotient $Q(P) = |\mathcal{N}(P)|/L$ of the normalized optimal fiber P in its active lattice L . This paper determines how flexible that finite-fiber model becomes beyond active rank two.

First, we prove a converse realization theorem: up to affine lattice equivalence, the normalized optimal fibers of completion-total integer counting systems are exactly the finite lattice-convex configurations. The construction simultaneously enforces nonnegativity, constant optimal length, saturation of the active difference lattice, and a strictly positive kernel direction. Thus higher-rank completion topology can be studied directly as finite lattice-convex translation geometry.

We then develop two rank-raising operations. A minimal active-rank- s fiber with $s+1$ points yields $Q(P) \simeq \bigvee^s S^1$ and fundamental group F_s , while the Boolean cube $\{0,1\}^s$ yields T^s . More generally, for finite lattice-convex configurations $P \subset L$ and $R \subset M$ we prove the product law

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

Its interval factor gives a prism operation $Q(P \times \{0,1\}) \simeq Q(P) \times S^1$. In contrast, a translation-isolated apex extension raises active rank by one but satisfies

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Hence one-rank extensions naturally split into direct-product and free-product behavior on fundamental groups.

Finally, combining the product theorem with an explicit rank-two seed family gives, for every fixed active rank $s \geq 2$, completion-total systems with

$$\pi_1(Q_{m,s}) \cong \mathbb{Z}^s$$

while every higher Betti number β_j , $2 \leq j \leq s$, is unbounded. Their exact Betti polynomial is

$$(1 + 2t + (m-1)t^2)(1+t)^{s-2}.$$

Thus neither active rank nor the abstract fundamental group controls higher completion topology. The contribution is specific to optimal-completion fibers and their periodic translation quotients; the polyhedral, nerve, poset-homotopy, and lattice-ideal tools used in the proofs are classical.

Keywords. deterministic pushdown normalization; shortest completion; optimal Parikh fiber; lattice-convex set; periodic nerve; lattice ideal; homotopy type; Betti numbers; rank-raising topology.

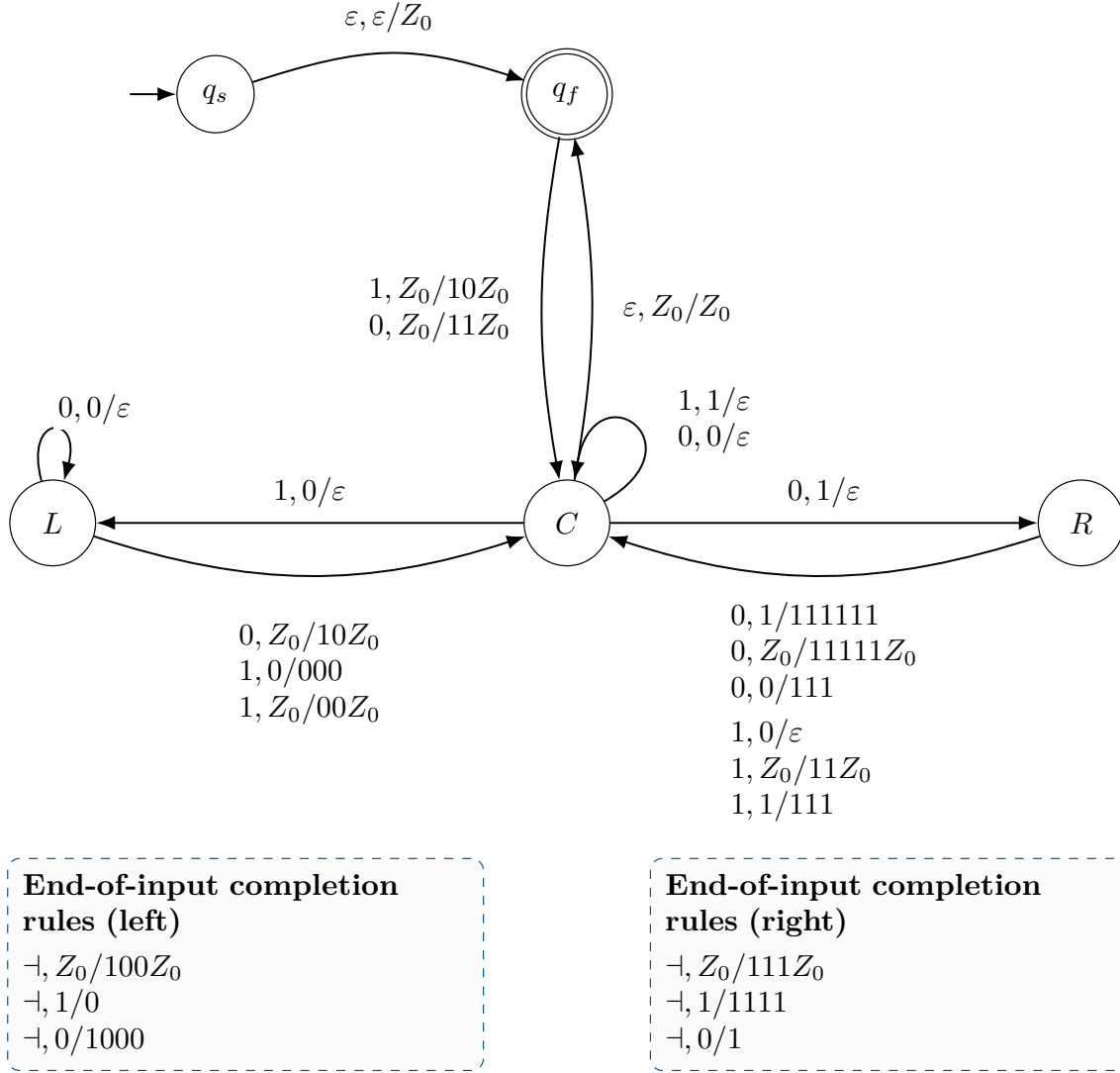


Figure 1: The completion-aware 2:1 DPDA that provides the operational provenance of the completion program. Solid arcs process binary input; the blue dashed boxes mark right-endmarker completion rules. All textual labels remain black. The figure is motivational: none of the higher-rank topological theorems uses this transition table as a hypothesis.

1 Introduction

The topology considered here comes from a normalization problem, not from an abstract attempt to manufacture cell complexes. The starting point is the completion-aware 2:1 deterministic pushdown machine in Figure 1. Its finite control and literal stack jointly encode how a prefix must be completed to restore weighted balance. Passing from the scalar 2:1 residual to general coprime $p:q$ balance isolates the signed residual $qN_1 - pN_0$; passing further to an integer matrix counting system replaces a unique shortest completion demand by a finite family of equally short nonnegative completion vectors. The geometry of that optimal fiber is the source of the topology studied below.

For $A \in \mathbb{Z}^{k \times d}$ and a residual r for which a nonnegative completion exists, define

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{Z}_{\geq 0}^d, Ax = -r\}, \quad \mathcal{O}_A(r) = \{x \in \mathbb{Z}_{\geq 0}^d : Ax = -r, \mathbf{1}^\top x = \ell_A(r)\}.$$

Fix $x_0 \in \mathcal{O}_A(r)$ and put

$$L_A(r) = \langle \mathcal{O}_A(r) - \mathcal{O}_A(r) \rangle_{\mathbb{Z}}, \quad P_A(r) = \mathcal{O}_A(r) - x_0.$$

The rank of $L_A(r)$ is the *active rank*. The filled shortest-prefix quotient is determined, up to homotopy, by the finite translation geometry of $P_A(r)$. For the rank-raising results below we need only one rank-two homological fact: there is no integral homology in degree at least three, and H_2 is torsion-free. We include a proof of precisely that fact in Theorem 3.7; no complete rank-two homotopy classification is used as a black box.

The present paper begins from this finite-fiber model. Its first question is more basic than a rank-three case split: *which finite configurations can occur as optimal fibers at all?* Once that realization question is answered, higher-rank topology becomes a problem about finite lattice-convex sets and their periodic translation nerves.

Main results. The contributions are as follows.

- (i) Every finite lattice-convex configuration is realized, up to affine lattice equivalence, as the normalized optimal fiber of a completion-total integer counting system. Together with the known necessity of lattice-convexity, this gives an exact realization equivalence.
- (ii) If an active-rank- s fiber has the minimum possible size $s + 1$, then its quotient is $\bigvee^s S^1$ and its fundamental group is F_s . An explicit scalar family realizes this branch in every rank.
- (iii) Periodic optimal-fiber quotients satisfy a product theorem:

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

In particular, Boolean cubes realize all higher tori, and a prism factor raises rank by one via product with S^1 .

- (iv) A different one-rank extension, the translation-isolated apex, satisfies

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Thus apex and prism extensions produce free product with $\mathbb{Z}$ and direct product with $\mathbb{Z}$, respectively, at the level of fundamental groups.

- (v) For every fixed $s \geq 2$ there are completion-total systems of active rank s with fixed fundamental group $\mathbb{Z}^s$ and unbounded β_j for every $2 \leq j \leq s$. Their Betti numbers are explicit.
- (vi) The abstract fundamental group does not determine the active displacement information: an F_3 quotient occurs both with active rank two and with active rank three, but the linear displacement kernel is different.

Scope and novelty boundary. Slack-variable and standard-form encodings of lattice polytopes are classical, as are nerve theorems, poset homotopy, Künneth formulas, lattice ideals, and cellular resolutions. We do not claim new general results in those subjects. The completion-specific contribution is the exact realization of lattice-convex sets as *minimum-length completion fibers*, together with the apex/product calculus for their periodic translation quotients and its fixed-rank consequences. A targeted literature search located adjacent polyhedral and homotopical tools but no direct prior theorem matching this complete chain. The novelty language below is therefore deliberately package- and specialization-level.

2 The periodic optimal-fiber model

We work intrinsically with a finite lattice-convex configuration and then return to counting systems through the realization theorem.

Definition 2.1 (Lattice-convex configuration). Let L be a lattice of rank s . A finite set $P \subset L$ is *lattice-convex* if

$$P = \text{conv}(P) \cap L.$$

It *spans* L if $L = \langle P - P \rangle_{\mathbb{Z}}$.

After translation we always assume $0 \in P$.

Definition 2.2 (Periodic nerve and translation quotient). For a finite $P \subset L$, let $\mathcal{N}(P)$ be the simplicial complex with vertex set L in which a finite set $F \subset L$ spans a simplex exactly when there exists $c \in L$ such that

$$c - F \subseteq P.$$

Equivalently, $\mathcal{N}(P)$ is the nerve of the family of translates $\{\lambda + P : \lambda \in L\}$. Translation by L acts freely on vertices and simplices. Put

$$Q(P) = |\mathcal{N}(P)|/L.$$

The quotient has one vertex orbit. Its edge orbits are indexed by differences in $P - P$, and its higher faces record simultaneous translate intersections.

Proposition 2.3 (Completion baseline). *For every completable residual of an integer counting system, the normalized optimal fiber $P_A(r)$ is finite and lattice-convex in $L_A(r)$. The filled shortest-prefix quotient $I_A(r)$ of the completion geometry satisfies*

$$I_A(r) \simeq Q(P_A(r)).$$

Proof. If $x_0 \in \mathcal{O}_A(r)$ and $L = L_A(r)$, every point of $(x_0 + L) \cap \mathbb{Z}_{\geq 0}^d$ has the same image under A and the same coordinate sum as x_0 , hence is also optimal. Therefore

$$\mathcal{O}_A(r) = (x_0 + L) \cap \mathbb{Z}_{\geq 0}^d,$$

which immediately gives $P_A(r) = \text{conv}(P_A(r)) \cap L$.

For the topological reduction, saturate the cubical shortest-prefix region by all L -translates. Any nonempty finite intersection of translates is a coordinate down-set after translation and is therefore contractible. The equivariant nerve theorem identifies the saturation with the nerve of this cover. A finite family of translates intersects exactly when one common optimal endpoint completes all of them, which is precisely the simplex criterion in Theorem 2.2. Passing to the free translation quotient yields the displayed equivalence. This is the only completion-to-periodic-nerve reduction needed below. $\square$

The finite presentation also supplies a canonical displacement map.

Proposition 2.4 (Difference presentation and displacement). *Let $P \subset L$ be finite, spanning, and normalized by $0 \in P$. Then*

$$\pi_1(Q(P)) \cong \left\langle X_p \ (p \in P), \ X_0 = 1 \mid X_p^{-1} X_q = X_s^{-1} X_t \text{ whenever } q - p = t - s \right\rangle. \quad (1)$$

Moreover there is a canonical displacement homomorphism

$$\delta : \pi_1(Q(P)) \longrightarrow L, \quad \delta(X_p) = p. \quad (2)$$

Proof. In $\mathcal{N}(P)/L$, oriented edge orbits are indexed by differences $d = q - p \in P - P$. Write the corresponding edge generator as a_d , with $a_{-d} = a_d^{-1}$. Every triple $p, q, s \in P$ supplies the triangular relation

$$a_{q-p} a_{s-q} = a_{s-p}.$$

Set $X_p = a_p$ for $p \in P$ and $X_0 = 1$. The triangle on $0, p, q$ gives

$$a_{q-p} = X_p^{-1} X_q.$$

Two edges represent the same edge orbit exactly when their differences agree, which gives precisely the relations in (1). Higher simplices do not affect the fundamental group.

Finally, sending the edge orbit a_d to its lattice difference d respects every triangular relation. It therefore defines the canonical displacement homomorphism δ , and $\delta(X_p) = p$. It is surjective because $L = \langle P - P \rangle = \langle P \rangle$ when $0 \in P$. $\square$

On first homology we denote the kernel by

$$K_{\text{lin}}(P) = \ker(H_1(Q(P); \mathbb{Z}) \rightarrow L).$$

3 Every lattice-convex configuration is an optimal fiber

The necessity in Theorem 2.3 has a converse.

Definition 3.1 (Completion-total system). An integer counting matrix A is *completion-total on its integer image* if every $b \in A\mathbb{Z}^d$ has a nonnegative preimage. A sufficient condition is the existence of a strictly positive vector $h \in \ker_{\mathbb{Z}} A$: any integer solution can then be translated by a large nonnegative multiple of h .

Theorem 3.2 (Lattice-convex realization). *Let P be a finite lattice-convex configuration spanning a rank- s lattice L . Then there exist an integer matrix A , a completable residual r , and a strictly positive $h \in \ker_{\mathbb{Z}} A$ such that the normalized optimal fiber $P_A(r)$ is affinely lattice-isomorphic to P and its active lattice has rank exactly s .*

Proof. Identify L with $\mathbb{Z}^s$ and translate so that $0 \in P$. Since $\text{conv}(P)$ is a lattice polytope, choose integral affine facet functions

$$\ell_j(z) = b_j - a_j^\top z \quad (1 \leq j \leq m)$$

such that

$$\text{conv}(P) = \{z \in \mathbb{R}^s : \ell_j(z) \geq 0 \ \forall j\}.$$

Choose $B \in \mathbb{Z}^s$ so large coordinatewise that $z + B \in \mathbb{Z}_{\geq 0}^s$ for every $z \in P$. Define the integral affine embedding

$$E_0(z) = (z + B, \ell_1(z), \dots, \ell_m(z)) \in \mathbb{Z}^{s+m}.$$

For lattice points $z \in \mathbb{Z}^s$ we have

$$E_0(z) \geq 0 \iff z \in P.$$

Indeed, points of P satisfy every facet inequality and the chosen coordinate shifts; a lattice point outside $P = \text{conv}(P) \cap \mathbb{Z}^s$ violates at least one facet inequality.

Choose an integer

$$C \geq \max_{z \in P} \mathbf{1}^\top E_0(z)$$

and append one balancing coordinate:

$$E(z) = (E_0(z), C - \mathbf{1}^\top E_0(z)) \in \mathbb{Z}^n, \quad n = s + m + 1.$$

Then

$$E(z) \geq 0 \iff z \in P, \quad \mathbf{1}^\top E(z) = C \quad (z \in \mathbb{Z}^s). \quad (3)$$

Let

$$D = \{E(z) - E(0) : z \in \mathbb{Z}^s\} \subset \mathbb{Z}^n.$$

The first s coordinates of $E(z) - E(0)$ are exactly z , so projection onto those coordinates is a left inverse. Hence $D \cong \mathbb{Z}^s$ is a saturated direct summand of $\mathbb{Z}^n$. By (3), every $d \in D$ has coordinate sum zero.

Append one final coordinate and set

$$h = (1, \dots, 1) \in \mathbb{Z}^{n+1}, \quad K = (D \times \{0\}) + \mathbb{Z}h \subset \mathbb{Z}^{n+1}.$$

The last coordinate of h is one, so quotienting first by $\mathbb{Z}h$ identifies $\mathbb{Z}^{n+1}/\mathbb{Z}h$ with $\mathbb{Z}^n$; the image of K is the saturated lattice D . Thus K is saturated. Consequently there is an integer matrix A with

$$\ker_{\mathbb{Z}} A = K.$$

The vector h is strictly positive, so A is completion-total on its integer image.

Put

$$x_0 = (E(0), 0) \in \mathbb{Z}_{\geq 0}^{n+1}, \quad r = -Ax_0.$$

Any integer point in the fiber of Ax_0 has a unique expression

$$x = x_0 + (d, 0) + th, \quad d \in D, \quad t \in \mathbb{Z}.$$

Its last coordinate is t , so nonnegativity forces $t \geq 0$. Writing $d = E(z) - E(0)$ gives

$$\mathbf{1}^\top x = C + t(n+1).$$

Hence every minimum-length point has $t = 0$. At $t = 0$, nonnegativity is equivalent by (3) to $z \in P$. Therefore

$$\mathcal{O}_A(r) = \{(E(z), 0) : z \in P\}.$$

The active difference lattice is $D \times \{0\}$, which has rank s , and the map $z \mapsto (E(z), 0)$ is an affine lattice isomorphism from P onto the normalized optimal fiber. $\square$

Corollary 3.3 (Exact fiber characterization). *Up to affine lattice equivalence, the normalized optimal fibers of completion-total integer counting systems are exactly the finite lattice-convex configurations.*

Proof. Necessity is Theorem 2.3; sufficiency is Theorem 3.2. $\square$

3.1 The principal Taylor slice and the rank-two homological baseline

The next algebraic input is classical on the lattice-ideal side, but we record the precise principal slice selected by an optimal completion fiber. Let a completion realization have optimal fiber $\mathcal{O} \subset \mathbb{Z}_{\geq 0}^d$; after translating by a chosen basepoint, write P for its normalized fiber. Choose $x_0 \in \mathcal{O}$, and write $L = \langle \mathcal{O} - \mathcal{O} \rangle_{\mathbb{Z}}$. Over a field K , put $S = K[x_1, \dots, x_d]$ and

$$I_L = \langle x^{u^+} - x^{u^-} : u \in L \rangle.$$

Give S the $\mathbb{Z}^d/L$ -grading and let $\alpha = x_0 + L$. The monomial fiber of α has exponent set exactly $\mathcal{O}$. For a degree β , let $E_i(\beta)$ be the i -element subsets of its monomial fiber with monomial gcd one, and write $\beta \preceq \alpha$ when $\alpha - \beta$ is a monomial degree.

Lemma 3.4 (Optimal-fiber Taylor slice). *For every $j \geq 0$,*

$$C_j(Q(P); K) \cong \bigoplus_{\beta \preceq \alpha} KE_{j+1}(\beta),$$

and under this basis correspondence the cellular boundary is the Bayer–Sturmfels Taylor differential specialized by $x_i \mapsto 1$.

Proof. Represent a j -simplex orbit by a $(j+1)$ -element subset $F \subset \mathcal{O}$. Subtract its coordinatewise minimum g ; the result is a gcd-one subset in a degree $\beta \preceq \alpha$. Translating F by an element of L translates its common gcd by the same amount, so the normalized gcd-one set depends only on the simplex orbit. Conversely, multiply a gcd-one set in degree $\beta \preceq \alpha$ by a monomial raising its degree to α . A different choice differs by an L -translation and gives the same orbit. These constructions are inverse.

When one vertex is deleted, the remaining set acquires a common monomial gcd. The Bayer–Sturmfels Taylor differential records that gcd as its coefficient; after $x_i \mapsto 1$, the coefficient becomes one and the signed term is precisely the oriented simplicial boundary [2]. $\square$

Lemma 3.5 (Principal truncation). *A homogeneous chain-homotopy equivalence between the lattice Taylor resolution and a minimal $\mathbb{Z}^d/L$ -graded free resolution restricts to the principal degree down-set $\{\beta : \beta \preceq \alpha\}$.*

Proof. Comparison maps and homotopies may be chosen homogeneous of degree zero. A homogeneous component from degree β to degree γ can be nonzero only when $\gamma \preceq \beta$. Hence every comparison map and homotopy preserves the principal down-set. $\square$

Lemma 3.6 (Rank-two torus-point injectivity). *If $\text{rank } L = 2$, then over every field K the third differential of a minimal graded free resolution of S/I_L , after specialization at any torus character of L , is injective.*

Proof. Extend scalars to an algebraic closure and represent the character by a torus point ξ . For a nontrivial character, some lattice binomial is nonzero at ξ , so ξ lies outside the support of S/I_L and the specialized free resolution is acyclic. For the trivial character, localize at the algebraic torus. If g_1, g_2 is a lattice basis, the Laurent lattice ideal is generated by $x^{g_1} - 1$ and $x^{g_2} - 1$, a regular sequence of length two in a regular Laurent ring, so Tor vanishes above degree two. Peeva–Sturmfels prove $\text{pd}(S/I_L) \leq 3$ in rank two [5]; there is no fourth term, hence vanishing of third homology is exactly injectivity of the third differential. $\square$

Proposition 3.7 (Rank-two homological baseline). *Let $P \subset L \cong \mathbb{Z}^2$ be finite, spanning, and lattice-convex. Then*

$$H_j(Q(P); \mathbb{Z}) = 0 \quad (j \geq 3),$$

and $H_2(Q(P); \mathbb{Z})$ is free abelian.

Proof. By Theorem 3.2, realize P as a completion-total optimal fiber without changing its active lattice or periodic quotient. Over every field, Theorems 3.4 and 3.5 identify the cellular chain complex, up to chain homotopy, with the principal slice of a minimal resolution of length at most three. By Theorem 3.6, its third differential is injective after specialization and remains injective after restriction to the principal down-set. Thus $H_j(Q(P); K) = 0$ for all fields K and all $j \geq 3$.

Since $Q(P)$ is finite, the universal coefficient theorem over $\mathbb{Q}$ and every prime field yields integral vanishing in degrees at least three. In degree three, the mod- ℓ universal-coefficient sequence gives

$$\text{Tor}_1^{\mathbb{Z}}(H_2(Q(P); \mathbb{Z}), \mathbb{F}_\ell) = 0$$

for every prime ℓ . Hence $H_2(Q(P); \mathbb{Z})$ is torsion-free and therefore free abelian. $\square$

Remark 3.8 (Coarse all-rank ceiling). The general Peeva–Sturmfels estimate $\text{pd}(S/I_L) \leq 2^{\text{rank } L} - 1$ gives only the coarse baseline vanishing $H_j(Q(P)) = 0$ for $j \geq 2^{\text{rank } L}$. No main theorem uses this coarse bound.

4 Minimal fibers and free groups in every rank

A spanning rank- s configuration has at least $s + 1$ points. At equality the quotient is completely rigid.

Theorem 4.1 (Minimal active fiber). *Let $P \subset L$ be a normalized, spanning, lattice-convex configuration with $\text{rank } L = s$ and $|P| = s + 1$. Then*

$$Q(P) \simeq \bigvee^s S^1, \quad \pi_1(Q(P)) \cong F_s, \quad H_j(Q(P); \mathbb{Z}) = 0 \quad (j \geq 2).$$

Moreover the abelianized displacement map

$$H_1(Q(P); \mathbb{Z}) \longrightarrow L$$

is an isomorphism, so $K_{\text{lin}}(P) = 0$.

Proof. The $s + 1$ points are affinely independent. No positive-dimensional subset can be a nontrivial translate of another subset of P : such a translation would give either a repeated nonzero difference among four vertices or three collinear vertices, in either case an affine dependence. Thus the only identifications in a fundamental simplex are the singleton vertices, which form one translation orbit. Consequently

$$Q(P) \cong \Delta^s / P^{(0)}.$$

Collapsing the $s + 1$ vertices of a contractible simplex to one point gives the homotopy cofiber of the inclusion of a discrete $(s + 1)$ -point set into Δ^s , hence a wedge of s circles.

After translating one point to 0, the remaining s points generate L by definition. Their affine independence makes them linearly independent, so they form a $\mathbb{Z}$ -basis of L . The s circle generators map to this basis under displacement, proving the final assertion. $\square$

Proposition 4.2 (Explicit scalar realization). *For every $s \geq 1$, the scalar weight vector*

$$\omega_s = (\underbrace{1, \dots, 1}_{s+1 \text{ entries}}, -(s+1))$$

with residual $r = -1$ has minimum completion length one and exactly the first $s + 1$ unit vectors as optimal completions. Its active rank is s , and the all-ones vector is a positive kernel vector.

Proof. The equation $\omega_s x = 1$ has the $s + 1$ positive unit vectors as length-one solutions; the final coordinate cannot occur in a length-one solution because its weight is negative. Thus the minimum is one and the displayed unit vectors are the complete optimum. Their difference lattice has rank s , while the sum of all $s + 2$ columns is zero, so the all-ones vector lies in the kernel. $\square$

At $s = 3$ this gives a genuine active-rank-three quotient with $\pi_1 \cong F_3$ and $K_{\text{lin}} = 0$.

5 Products and higher tori

The periodic nerve is compatible with Cartesian products in a form that is especially useful for raising active rank.

Theorem 5.1 (Optimal-fiber product theorem). *Let $P \subset L$ and $R \subset M$ be finite lattice-convex configurations spanning their lattices. Then $P \times R$ is lattice-convex in $L \oplus M$, has active rank $\text{rank } L + \text{rank } M$, and*

$$Q(P \times R) \simeq Q(P) \times Q(R).$$

The equivalence is $(L \oplus M)$ -equivariant before passage to the quotient.

Proof. Write $K = \mathcal{N}(P)$ and $H = \mathcal{N}(R)$. A finite set $F \subset L \times M$ is a simplex of $\mathcal{N}(P \times R)$ exactly when some $(c, d) \in L \times M$ satisfies

$$(c, d) - F \subset P \times R.$$

This is equivalent to

$$c - \pi_L(F) \subset P, \quad d - \pi_M(F) \subset R.$$

Hence $\mathcal{N}(P \times R)$ is the simplicial categorical product whose simplices are precisely the finite sets whose two coordinate projections are simplices of K and H .

On the posets of nonempty faces define

$$\phi(F) = (\pi_L F, \pi_M F), \quad \psi(A, B) = A \times B.$$

Then $\phi\psi = \text{id}$ and

$$F \subseteq \psi\phi(F)$$

for every face F . Pointwise comparable poset maps induce homotopic maps on order complexes [6]. Therefore ϕ and ψ induce mutually inverse homotopy equivalences after barycentric subdivision. The order complex of the product face poset is the standard barycentric triangulation of the product of the two barycentric subdivisions, so its realization is naturally homeomorphic to $|K| \times |H|$. The construction and the comparison $F \subseteq \psi\phi(F)$ commute with translations, so the equivalence is $(L \oplus M)$ -equivariant. Quotienting gives the asserted product law.

Finally,

$$\text{conv}(P \times R) = \text{conv}(P) \times \text{conv}(R),$$

so

$$(\text{conv}(P \times R)) \cap (L \times M) = (\text{conv}(P) \cap L) \times (\text{conv}(R) \cap M) = P \times R.$$

The difference lattice is $\langle P - P \rangle \oplus \langle R - R \rangle$, proving the rank statement. $\square$

Corollary 5.2 (Iterated products). *For finitely many spanning lattice-convex configurations $P_i \subset L_i$,*

$$Q\left(\prod_i P_i\right) \simeq \prod_i Q(P_i).$$

Every such product is realized as a genuine completion-total optimal fiber by Theorem 3.2.

The most symmetric instance is the Boolean cube.

Theorem 5.3 (Higher-torus realization). *For every $s \geq 1$ there is a completion-total integer counting system with active rank s and normalized optimal fiber affinely lattice-isomorphic to $\{0, 1\}^s$. Its translation quotient is*

$$Q(\{0, 1\}^s) \simeq T^s,$$

so

$$H_j(Q; \mathbb{Z}) \cong \mathbb{Z}^{\binom{s}{j}} \quad (0 \leq j \leq s).$$

Proof. The topological statement follows immediately from Theorem 5.1: $\{0, 1\}^s$ is the product of s rank-one intervals, and $Q(\{0, 1\}) \simeq S^1$ by Theorem 4.1. Thus its quotient is T^s .

For a direct completion-total realization, use $2s+1$ letters grouped into pairs (a_i, b_i) , $1 \leq i \leq s$, together with a padding letter c . Let A_s have s rows. For $1 \leq i < s$, row i has coefficient 1 on a_i, b_i , coefficient -1 on a_s, b_s , and 0 elsewhere. The final row has coefficient 1 on the first $2s$ letters and coefficient $-2s$ on c .

Take residual

$$r = (0, \dots, 0, -s),$$

so the completion equation $A_s x = -r$ has last coordinate s . If $t_i = x_{a_i} + x_{b_i}$ and $z = x_c$, the first $s - 1$ equations force

$$t_1 = \cdots = t_s = t,$$

while the last equation gives

$$st - 2sz = s, \quad t = 1 + 2z.$$

The total length is

$$st + z = s + (2s + 1)z,$$

so it is minimized exactly at $z = 0, t = 1$. Hence an optimal completion chooses exactly one letter from each pair, giving 2^s optima and normalized cube fiber. The within-pair swaps generate an active lattice of rank s . The all-ones vector is a strictly positive kernel vector because every difference row has total coefficient zero and the last row has $2s - 2s = 0$. $\square$

Corollary 5.4 (First third homology). *Active rank three admits*

$$Q \simeq T^3, \quad (\beta_0, \beta_1, \beta_2, \beta_3) = (1, 3, 3, 1).$$

This is the first possible active rank for nonzero third homology, because Theorem 3.7 gives $H_j = 0$ for $j \geq 3$ in active rank two.

6 Two one-rank extensions: apex versus prism

Product with the interval is one natural one-rank extension. A second extension has a different homotopy effect.

Theorem 6.1 (Translation-isolated apex). *Let $P \subset L$ be finite, spanning, and lattice-convex. In $L \oplus \mathbb{Z}e$ put*

$$P^+ = (P \times \{0\}) \cup \{e\},$$

where e has last coordinate one. Then P^+ is lattice-convex, has active rank $\text{rank } L + 1$, and

$$Q(P^+) \simeq Q(P) \vee S^1.$$

Proof. A lattice point of the pyramid $\text{conv}(P^+)$ has last coordinate 0 or 1. At height zero the slice is $\text{conv}(P)$, whose lattice points are exactly P ; at height one the slice is only the apex. Thus P^+ is lattice-convex, and its difference lattice is $L \oplus \mathbb{Z}e$.

No positive-dimensional face containing the apex has a nontrivial translate inside P^+ . Indeed, a translation with last coordinate zero must fix the unique height-one point and is therefore zero; a translation with last coordinate ± 1 sends one of the two height levels outside $\{0, 1\}$. Hence every nontrivial face identification is confined to the base simplex, although all vertices, including the apex, belong to one translation orbit.

Choose the simplex of the periodic nerve corresponding to the translate P^+ itself as a fundamental simplex; it is the cone on the base simplex $\Delta(P)$. Every simplex orbit has a representative in this cone. The preceding height argument has two consequences for its face identifications. First, all identifications between positive-dimensional base faces are exactly those already defining $Q(P)$. Second, a positive-dimensional face containing the apex cannot be identified with a distinct face containing the apex: a translating vector with last coordinate zero would have to fix the unique height-one vertex and hence be zero, while a vector with nonzero last coordinate cannot preserve the two-level pattern of a mixed face. Only the apex vertex itself is identified with the common vertex orbit, because the translation action is transitive on vertices.

Consequently the quotient is obtained from the cone $C\Delta(P)$ by applying the old base quotient

$$q : \Delta(P) \longrightarrow Q(P)$$

and identifying the cone point with the vertex orbit of $Q(P)$. Equivalently it is the homotopy coequalizer of q and the constant map $\Delta(P) \rightarrow Q(P)$ at that vertex. Since $\Delta(P)$ is nonempty and contractible, q is homotopic to the constant map. The homotopy coequalizer of two homotopic maps from a nonempty contractible space is homotopy equivalent to $Q(P) \vee S^1$ (for example by the standard mapping-cylinder model). This proves the theorem. $\square$

Corollary 6.2 (Prism law). *For every finite spanning lattice-convex P ,*

$$Q(P \times \{0, 1\}) \simeq Q(P) \times S^1.$$

Proof. Apply Theorem 5.1 and $Q(\{0, 1\}) \simeq S^1$. $\square$

Thus apex and prism both add one active lattice direction but act differently:

$$\pi_1(Q(P^+)) \cong \pi_1(Q(P)) * \mathbb{Z}, \quad \pi_1(Q(P \times \{0, 1\})) \cong \pi_1(Q(P)) \times \mathbb{Z}.$$

For torsion-free homology, the Künneth formula also gives

$$H_j(Q(P) \times S^1) \cong H_j(Q(P)) \oplus H_{j-1}(Q(P)),$$

while apex extension preserves H_j for $j \geq 2$.

Corollary 6.3 (Rank-three branches). *Both operations produce genuine completion-total active-rank-three families.*

- (a) *Apexing a rank-two F_2 fiber gives F_3 ; apexing a rank-two F_3 fiber gives F_4 .*
- (b) *Apexing a rank-two toral fiber gives fundamental group $\mathbb{Z}^2 * \mathbb{Z}$ and, for the basic torus, homotopy type $T^2 \vee S^1$.*
- (c) *Prisming the rank-two minimal F_2 branch gives an aspherical $F_2 \times \mathbb{Z}$ quotient with Betti vector $(1, 3, 2)$. Prisming a rank-two F_3 wedge branch gives an aspherical $F_3 \times \mathbb{Z}$ quotient with Betti vector $(1, 4, 3)$.*

7 Fixed-rank higher homology and displacement

We now combine the rank-raising calculus with an explicit rank-two seed.

Proposition 7.1 (Rank-two seed family). *For every $m \geq 2$ there is a completion-total scalar system with active rank two and a normalized fiber affinely equivalent to*

$$P_m = \{(t, 0) : 0 \leq t \leq m\} \cup \{(0, 1), (m-1, -1)\} \subset \mathbb{Z}^2. \quad (4)$$

Its quotient satisfies

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2, \quad H_1(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^2, \quad H_2(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^{m-1},$$

and has no homology above degree two. Equivalently, its Betti polynomial is

$$1 + 2t + (m-1)t^2.$$

Proof. Take $\alpha = 2m + 1$ and the four scalar weights

$$\omega_m = (1 - m\alpha, 1 - (m-1)\alpha, 1, 1 + \alpha),$$

with residual

$$r_m = \alpha m(m-1) - 2m.$$

The strictly positive vector

$$h_m = (1, 1, (2m - 2)\alpha - 3, 1)$$

lies in the kernel. All four weights are 1 modulo α , so every completion length satisfies

$$n \equiv -r_m \equiv 2m \pmod{2m + 1}.$$

At length $2m$, the complete optimum is

$$A_t = (t, m - t, m - t, t) \quad (0 \leq t \leq m),$$

$$B = (0, m + 1, 0, m - 1), \quad C = (m - 1, 0, m + 1, 0).$$

Indeed, eliminating the common congruence term gives

$$x_2 + mx_3 + (m + 1)x_4 = m(m + 1),$$

and reduction modulo m forces $x_2 + x_4 \in \{0, m, 2m\}$, yielding exactly C , the A_t , and B .

The equal-length kernel has basis

$$g = (1, -1, -1, 1), \quad h = (0, 1, -m, m - 1),$$

and relative to A_0 the optimal points have coordinates

$$A_t - A_0 = tg, \quad B - A_0 = h, \quad C - A_0 = (m - 1)g - h,$$

which is (4). Its convex hull is the lattice quadrilateral

$$(0, 0), (m - 1, -1), (m, 0), (0, 1).$$

All four boundary edges are primitive. The Euclidean area is m , so Pick's theorem gives $m - 1$ interior lattice points. These are exactly $(1, 0), \dots, (m - 1, 0)$; together with the four boundary vertices this is precisely P_m . Hence P_m is lattice-convex.

We next compute the fundamental group directly from the difference presentation (1), without invoking a general planar classification. Write

$$x = X_{(1,0)}, \quad y = X_{(0,1)}, \quad z = X_{(m-1,-1)}.$$

Repeated horizontal differences give $X_{(t,0)} = x^t$. The repeated difference $(0, 1)$ between the pairs $((0, 0), (0, 1))$ and $((m - 1, -1), (m - 1, 0))$ gives

$$y = z^{-1}x^{m-1}, \quad z = x^{m-1}y^{-1}.$$

For $1 \leq t \leq m - 1$, the repeated difference $(-t, 1)$ between

$$((t, 0), (0, 1)) \quad \text{and} \quad ((m - 1, -1), (m - 1 - t, 0))$$

gives

$$x^{-t}y = yx^{-t}.$$

Taking $t = 1$ shows $xy = yx$. Thus the presentation is generated by the commuting pair x, y . The displacement map (2) sends them to the standard basis of $\mathbb{Z}^2$, so it rules out any further relation and yields

$$\pi_1(Q(P_m)) \cong \mathbb{Z}^2.$$

In particular $H_1(Q(P_m); \mathbb{Z}) \cong \mathbb{Z}^2$. The rank-two homological baseline Theorem 3.7 gives no homology above degree two and makes H_2 free abelian.

It remains to compute the Euler characteristic by simplex orbits. Put $H = \{(t, 0) : 0 \leq t \leq m\}$. A horizontal-only orbit can be normalized to have minimum horizontal coordinate zero, so its alternating contribution is

$$\sum_{k \geq 1} (-1)^{k-1} \binom{m}{k-1} = 0.$$

For either fixed off-line point, faces containing that point and a nonempty horizontal subset contribute

$$\sum_{k=1}^{m+1} (-1)^k \binom{m+1}{k} = -1,$$

while faces containing both off-line points have total alternating contribution

$$\sum_{k=0}^{m+1} (-1)^{k+1} \binom{m+1}{k} = 0.$$

The only cross-identifications between the two one-off-line families are the m edge pairs

$$\{(0, 1), (t, 0)\} \sim \{(m-1, -1), (m-1-t, 0)\}, \quad 0 \leq t \leq m-1.$$

A nonzero vertical translation can match at most one horizontal point because each off-line level contains only one point, so there is no higher-dimensional cross-identification. Identifying each duplicate edge orbit raises the alternating count by one. Therefore

$$\chi(Q(P_m)) = 0 - 1 - 1 + m = m - 2.$$

Since $\beta_0 = 1$, $\beta_1 = 2$, and higher homology vanishes,

$$m - 2 = 1 - 2 + \beta_2,$$

so $\beta_2 = m - 1$. Torsion-freeness has already been supplied by Theorem 3.7. $\square$

Corollary 7.2 (Apex branch with unbounded second homology). *Let P_m^+ be the apex extension of the seed in Theorem 7.1. Then P_m^+ has active rank three and*

$$Q(P_m^+) \simeq Q(P_m) \vee S^1, \quad \pi_1(Q(P_m^+)) \cong \mathbb{Z}^2 * \mathbb{Z}.$$

Moreover

$$H_2(Q(P_m^+); \mathbb{Z}) \cong \mathbb{Z}^{m-1}, \quad H_j(Q(P_m^+); \mathbb{Z}) = 0 \quad (j \geq 3).$$

Hence β_2 is unbounded at fixed active rank three and fixed fundamental group $\mathbb{Z}^2 * \mathbb{Z}$.

Proof. Apply Theorem 6.1 to Theorem 7.1. Wedging on a circle changes only H_1 and takes the fundamental group to the free product with $\mathbb{Z}$. $\square$

The prism operation behaves differently: it moves the seed's second homology into degree three. This leads to the main all-rank phenomenon.

Theorem 7.3 (Fixed rank and fixed fundamental group with unbounded higher homology). *For every $s \geq 2$ there is a one-parameter family of completion-total integer counting systems whose normalized optimal fibers have active rank exactly s and whose quotients $Q_{m,s}$ satisfy*

$$\pi_1(Q_{m,s}) \cong \mathbb{Z}^s,$$

while every Betti number $\beta_j(Q_{m,s})$, $2 \leq j \leq s$, is unbounded as $m \rightarrow \infty$.

More precisely, take

$$P_{m,s} = P_m \times \{0, 1\}^{s-2}.$$

Then

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}$$

and its Betti polynomial is

$$\boxed{(1 + 2t + (m-1)t^2)(1+t)^{s-2}}. \quad (5)$$

Equivalently,

$$\boxed{\beta_j = \binom{s-2}{j} + 2\binom{s-2}{j-1} + (m-1)\binom{s-2}{j-2}}, \quad (6)$$

with out-of-range binomial coefficients interpreted as zero. In particular

$$\beta_1 = s, \quad \beta_s = m-1.$$

Proof. The active ranks add under products, so $P_{m,s}$ has rank $2 + (s-2) = s$. By Theorems 5.1 and 5.3,

$$Q(P_{m,s}) \simeq Q(P_m) \times T^{s-2}.$$

Its fundamental group is

$$\mathbb{Z}^2 \times \mathbb{Z}^{s-2} = \mathbb{Z}^s.$$

The two factors have torsion-free homology, so the integral Künneth formula has no Tor terms and multiplies Betti polynomials. This gives (5) and (6). For $2 \leq j \leq s$, the coefficient

$$\binom{s-2}{j-2}$$

of $m-1$ is positive, so every such β_j is unbounded. Finally, Theorem 3.2 realizes each product configuration as a completion-total optimal fiber. $\square$

Corollary 7.4 (Unbounded third homology in rank three). *At active rank three there are completion-total systems with*

$$\pi_1 \cong \mathbb{Z}^3, \quad (H_0, H_1, H_2, H_3) \cong (\mathbb{Z}, \mathbb{Z}^3, \mathbb{Z}^{m+1}, \mathbb{Z}^{m-1}).$$

Thus $\beta_3 = m-1$ is unbounded at fixed active rank and fixed fundamental group. The case $m=2$ is T^3 .

7.1 The displacement map contains information beyond π_1

The minimal rank-three fiber from Theorem 4.1 has

$$\pi_1 \cong F_3, \quad H_1 \cong \mathbb{Z}^3 \xrightarrow{\cong} L \cong \mathbb{Z}^3,$$

so $K_{\text{lin}} = 0$. For a rank-two comparison, take

$$P_F = \{(0,0), (0,1), (1,1), (3,2)\} \subset \mathbb{Z}^2.$$

Its convex hull is the triangle with vertices $(0,0), (0,1), (3,2)$ and normalized area 3; Pick's theorem shows that $(1,1)$ is its unique interior lattice point, so P_F is lattice-convex. The six unordered nonzero pair differences

$$(0,1), (1,1), (3,2), (1,0), (3,1), (2,1)$$

are distinct up to sign. Hence the presentation (1) has no nontrivial repeated-difference relation and is free on the three nonzero vertices:

$$\pi_1(Q(P_F)) \cong F_3.$$

But the displacement map on abelianization is a surjection

$$\mathbb{Z}^3 \longrightarrow \mathbb{Z}^2,$$

so its kernel is primitive of rank one. By Theorem 3.2, P_F also occurs as a genuine completion-total optimal fiber. We therefore obtain the following intrinsic conclusion.

Corollary 7.5 (Same fundamental group, different displacement visibility). *The abstract group $\pi_1(Q(P))$ does not determine the active rank or the linear zero-displacement subgroup $K_{\text{lin}}(P)$. In particular F_3 occurs with both $K_{\text{lin}} = 0$ and $K_{\text{lin}} \cong \mathbb{Z}$.*

This statement is purely about the intrinsic completion quotient. It does not assert a physical stack-carrier lower bound; such claims require a separately specified transport model.

8 Computational rank-three regression

The main results above are symbolic. Two finite audits were used only to test the quotient conventions. First, for the seed family P_m with $2 \leq m \leq 7$, every nonempty subset of P_m was grouped into its lattice-translation orbit, the integral simplicial boundary matrices of the finite quotient were constructed, and exact ranks were computed. In all six cases the calculation returned

$$\chi(Q(P_m)) = m - 2, \quad (\beta_0, \beta_1, \beta_2) = (1, 2, m - 1),$$

with all higher Betti numbers zero, exactly as in Theorem 7.1. This audit checks the simplex-orbit and Euler-characteristic bookkeeping but is not used in the proof.

Second, a finite Boolean audit was used to sample the first rank-three homological regimes.

All full-rank subsets of $\{0, 1\}^3$ were enumerated up to the 48 affine symmetries of the cube. There are 12 symmetry types: four with four points, three with five points, three with six points, one with seven points, and the full cube. Exact integer boundary matrices of the finite quotients produced the Betti profiles shown in Table 1. The table is regression evidence only and is not used to infer a complete rank-three classification.

Representative observed Betti profile	structural interpretation
(1, 3, 0, 0)	minimal rank-three free branch
(1, 4, 0, 0, 0)	free branch with one additional loop
(1, 3, 1, 0, 0)	first intermediate H_2 regime
(1, 3, 2, 0, ...)	intermediate H_2 regime
(1, 3, 3, 0, ...)	toral first/second homology without H_3
(1, 3, 4, 0, ...)	enlarged second homology
(1, 3, 3, 1, 0, ...)	full cube T^3

Table 1: Distinct Betti profiles observed in the Boolean rank-three regression atlas. The enumeration is a consistency check; none of the main theorems depends on it.

9 Related work and novelty boundary

Several ingredients used here have substantial independent literatures. Bayer and Sturmfels construct cellular resolutions for monomial modules and lattice ideals [2]. Peeva and Sturmfels give structural and projective-dimension results for lattice ideals [5]. Nerve and poset-fiber methods are classical; we use the equivariant nerve framework in the form of Fernós–Futer–Hagen [3] and the elementary comparable-map principle associated with Quillen’s poset homotopy

machinery [6]. Standard algebraic-topology facts about wedges, products, fundamental groups, and the Künneth theorem are used in their usual form [4].

On the polyhedral side, representing lattice-convex sets by systems of linear inequalities and passing to slack-variable standard forms is classical integer and polyhedral geometry. Modern work on relaxation complexity explicitly treats finite lattice-convex sets as the integer points of polyhedral relaxations [1]; Schrijver provides the standard linear/integer programming background [7]. Accordingly, Theorem 3.2 is not a claim to have invented slack variables. Its extra content is the completion-specific simultaneous enforcement of a constant optimal length, a saturated active difference lattice, and a strictly positive kernel direction, so that *exactly* the prescribed lattice points occur as the minimum-length completion fiber.

Likewise, Theorem 5.1 is not a new general theorem about products of nerves or products of posets. The contribution is the exact factorization of the periodic optimal-fiber nerve and the induced rank-raising calculus inside the completion model. The all-rank theorem Theorem 7.3 is a package-level consequence of that calculus: fixed active rank and fixed toral fundamental group coexist with unbounded higher homology.

A targeted search for direct overlap found adjacent results in polyhedral relaxations, cellular lattice-ideal resolutions, nerve theorems, and poset homotopy, but no theorem matching the complete specialization

completion-total realization $\longrightarrow$ periodic optimal-fiber quotient
 $\longrightarrow$ apex/product rank raising $\longrightarrow$ fixed rank and fixed π_1 with unbounded higher homology.

This is a negative search result rather than a priority proof, so the novelty claim is intentionally conservative.

10 Conclusion

The higher-rank completion problem is substantially more flexible than the rank-two classification suggests. The key foundational fact is an exact realization equivalence: finite lattice-convex configurations are not merely necessary shadows of optimal completion fibers; every such configuration can be realized by a completion-total integer counting system without changing its active lattice geometry.

Once this is known, rank raising becomes geometric. Minimal fibers give free groups in every rank, Boolean products give higher tori, and the general product theorem turns Cartesian products of fibers into products of translation quotients. The apex operation is complementary: it raises active rank by one while wedging on a circle. Thus two elementary lattice-convex extensions already distinguish free-product and direct-product behavior.

The resulting fixed-rank families show why active rank and fundamental group are insufficient invariants. For every $s \geq 2$, one may keep

$$\text{rank } L = s, \quad \pi_1 \cong \mathbb{Z}^s$$

fixed while making every higher Betti number β_j , $2 \leq j \leq s$, arbitrarily large. Separately, the same abstract group F_3 can occur with different active ranks and different displacement kernels. The topology is therefore controlled by the full translation geometry of the optimal fiber, not by the deck lattice or fundamental group alone.

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